

Satpuda Core ERP Transformation Master Guide

Documentation-only blueprint to transform the current medical store app into a complete ERP with Sales Return, Purchase Return, stronger calculation tables, ledger-based account clearing, GST reports, profit reports, polished UI, and complete exports. No application code is changed by this guide.

Generated April 25, 2026 after scanning the current project surface, database schema, export system, Python/Tkinter modules, React purchase-entry helper, and the previously generated accounting/table audit guides.

1. Current Project Surface

The current project is a desktop Python/Tkinter SQLite application with ttkbootstrap styling and a small React helper for web purchase entry/import. It already has useful ERP seeds: billing, purchases, inventory, histories, customers, suppliers, doctors, shelf/location management, import, backup, and exports.

Area	Current files / features	ERP transformation needed
Sales	ui/billing.py, widgets/bill_preview.py, widgets/bill_edit.py	Central calculation service, historical GST/cost snapshots, return support, payment allocation.
Purchase	ui/purchase.py, purchase-entry-web, ui/import_purchases.py	Supplier-linked batches, input GST lines, stock ledger, purchase returns.
Histories	ui/sales_history.py, ui/purchase_history.py	Ledger-based status, void/correction flows, return columns, ERP-grade reports.
Inventory	ui/inventory.py, ui/shelf_management.py	Medicine master + batch stock, stock ledger, valuation, supplier/batch traceability.
Accounts	sales/purchases due columns and triggers	Replace running-row due logic with party_ledger, payments, allocations.
Exports	core/export_manager.py, Settings exports, history exports	Add GST, returns, ledger, stock valuation, profit, audit exports; improve PDF/Excel layouts.

2. ERP Goal

The target ERP should treat every business event as posted documents plus ledger entries. Screens should preview calculations, but saves should call one shared engine that posts invoice rows, tax rows, stock ledger rows, party ledger rows, payment rows, and allocation rows inside one database transaction.

- No page should have its own independent GST/due/profit formula.
- Every sale, purchase, return, payment, stock adjustment, void, and correction must leave an audit trail.
- History should always remain clear because invoice status and party balance are derived from ledgers, not manually rewritten flags.
- GST and profit reports must come from stored historical item facts, not current medicine values.

3. ERP Module Map

Module	Pages / UI	Purpose
Dashboard	Home ERP dashboard	Today sales, collections, purchases, dues, low stock, expiry, GST payable estimate, pending returns.
Masters	Customers, Suppliers, Doctors, Medicine Master, Batch Master, Locations	Clean master data with active/inactive status and audit fields.
Sales	POS/Billing, Sales Return, Sales History, Customer Ledger	Invoice, collect payment, credit use, returns, print, export.
Purchase	Purchase Entry, Purchase Return, Purchase History, Supplier Ledger	Purchase stock, supplier payments, debit notes, input GST.
Inventory	Stock, Batch, Expiry, Movement, Valuation, Reorder	Stock ledger, batch traceability, supplier source, location control.
Accounts	Payments, Receipts, Allocations, Party Balances	Customer due/credit and supplier payable/advance.
GST	Sales GST, Purchase GST, Return GST, Net GST, HSN Summary	GST filing-ready reports.
Reports/Exports	CSV, Excel, PDF/HTML dashboards	Management, tax, stock, audit, profit, return reports.

Module	Pages / UI	Purpose
Admin	Backup, Settings, Numbering, Roles, Audit Log	ERP control and security.

4. Core Calculation Architecture

4.1 Shared Services

Service	Responsibility
money_service	Decimal parsing, half-up rounding, nearest rupee, formatting.
tax_service	GST inclusive/exclusive formulas, line GST split, discount allocation, HSN/rate summaries.
sales_service	Validate sale, calculate totals, save sale, post customer ledger, post stock out.
purchase_service	Validate purchase, calculate totals, save purchase, post supplier ledger, post stock in.
return_service	Sales return and purchase return with GST, stock, account, and profit reversal.
ledger_service	Party balance, payment posting, FIFO/manual allocations, invoice remaining due.
stock_service	Stock ledger posting, batch availability, valuation, reorder, expiry.
report_service	GST reports, profit, stock, due, payment, audit, return reports.

4.2 Money Rules

```

Use Decimal, not float.
money(x) = Decimal(x).quantize(Decimal("0.01"), ROUND_HALF_UP)
invoice_round_off = nearest_rupee(pre_round_total) - pre_round_total
grand_total = pre_round_total + invoice_round_off

```

- Store values in integer paise or Decimal-safe text/NUMERIC columns. Avoid SQLite REAL for new money fields where possible.
- Round item taxable and GST values to 2 decimals, then sum item values to header totals.

5. Target Database Tables

5.1 Master and Inventory Tables

Table	Essential columns
customers	id, name, phone, address, gstin, state_code, opening_balance, credit_limit, is_active, created_at, updated_at
suppliers	id, name, phone, address, gstin, state_code, dl_numbers, payment_terms_days, opening_balance, is_active, created_at, updated_at
medicine_master	id, name, type, manufacturer, content_drug, hsn_code, default_gst_rate, schedule, is_active, created_at, updated_at
medicine_batches	id, medicine_id, supplier_id, purchase_id, purchase_item_id, batch_no, expiry_date, unit_factor, stock_qty_available, cost_rate_ex_gst, cost_rate_inc_gst, mrp, location_id, created_at, updated_at
locations	id, rack, section, box, display_name, sort_order, is_active
stock_ledger	id, medicine_batch_id, source_type, source_id, qty_in, qty_out, cost_rate_ex_gst, balance_after, entry_date, note

Supplier should be linked at batch level, not only medicine name level, because one medicine can be purchased from many suppliers and batches.

5.2 Sales Tables

Table	Essential columns
sales	id, bill_no, customer_id, bill_date, supply_type, taxable_total, cgst_total, sgst_total, igst_total, gst_total, item_discount_total, bill_discount_type, bill_discount_input, bill_discount_amount, round_off, grand_total, invoice_status, payment_status, return_total, void fields, audit fields
sales_items	id, sale_id, medicine_batch_id, medicine_name_snapshot, batch_no, expiry_date, hsn_code, qty, unit_factor, rate_including_gst, gross_inclusive, item_discount_type/input/amount, bill_discount_allocated, taxable_value, gst_rate, cgst_amount, sgst_amount, igst_amount, gst_amount, line_total, cost_rate_ex_gst_snapshot, returned_qty
sale_returns	id, return_no, sale_id, customer_id, return_date, reason, taxable_total, gst_total, grand_total, refund_mode, status
sale_return_items	id, sale_return_id, sale_item_id, medicine_batch_id, return_qty, taxable_value, gst fields, line_total, cost_reversal, stock_ledger_id

5.3 Purchase Tables

Table	Essential columns
purchases	id, purchase_no, supplier_id, bill_number, supplier_bill_date, entry_date, rate_tax_mode, taxable_total, input_cgst_total, input_sgst_total, input_igst_total, input_gst_total, item_discount_total, bill_discount_type/input/amount, round_off, grand_total, invoice_status, payment_status, return_total, void/audit fields
purchase_items	id, purchase_id, medicine_id, created_batch_id, qty, free_qty, unit_factor, rate_ex_gst, gross_taxable_amount, discount fields, bill_discount_allocated, taxable_value, gst_rate, input GST fields, line_total, mrp, batch_no, expiry_date, stock_qty_added, returned_qty
purchase_returns	id, return_no, purchase_id, supplier_id, return_date, reason, taxable_total, input_gst_reversal_total, grand_total, settlement_mode, status
purchase_return_items	id, purchase_return_id, purchase_item_id, medicine_batch_id, return_qty, taxable_value, input GST reversal fields, line_total, stock_ledger_id

5.4 Accounts and Audit Tables

Table	Essential columns
party_ledger	id, party_type(customer/supplier), party_id, source_type, source_id, signed_amount, balance_after, entry_date, note
payments	id, party_type, party_id, invoice_type nullable, invoice_id nullable, method(cash/online/refund/adjustment/credit_use), amount, paid_at, reference_no
payment_allocations	id, payment_id/source_type/source_id, invoice_type, invoice_id, amount, allocated_at
invoice_sequences	id, document_type, financial_year, prefix, next_number
audit_log	id, user_id, action, table_name, row_id, old_json, new_json, created_at
app_users / roles	id, name, pin/password_hash, role, permissions_json, is_active

6. Exact GST and Discount Formulas

6.1 Purchase GST Exclusive

```
gross_taxable = qty * rate_ex_gst
item_discount_amount = discount_from_input(gross_taxable)
taxable_before_bill_discount = gross_taxable - item_discount_amount
allocated_bill_discount = bill_discount * taxable_before_bill_discount / total_taxable_before_bill_discount
taxable_value = taxable_before_bill_discount - allocated_bill_discount
gst_amount = taxable_value * gst_rate / 100
line_total = taxable_value + gst_amount
```

6.2 Sales GST Inclusive MRP

```
gross_inclusive = qty * sale_rate_including_gst
item_discount_amount = discount_from_input(gross_inclusive)
net_inclusive_before_bill_discount = gross_inclusive - item_discount_amount
allocated_bill_discount = bill_discount * net_inclusive_before_bill_discount / total_net_inclusive_before_bill_discount
net_inclusive = net_inclusive_before_bill_discount - allocated_bill_discount
taxable_value = net_inclusive * 100 / (100 + gst_rate)
gst_amount = net_inclusive - taxable_value
line_total = net_inclusive
```

6.3 GST Split

```
if supply_type == "intra_state":
    cgst = gst_amount / 2
    sgst = gst_amount / 2
    igst = 0
else:
    cgst = 0
    sgst = 0
    igst = gst_amount
```

7. Account Clear Logic

Account clear should be ledger-derived. Do not mark all old rows cleared/uncleared each time a new bill is created.

7.1 Customer

```
Sale invoice: party_ledger +grand_total
Customer payment: party_ledger -payment_amount
Sales return credit note: party_ledger -return_total
Customer refund: party_ledger +refund_amount
```

```
customer_balance = sum(customer_ledger_signed_amount)
> 0 = due, 0 = clear, < 0 = customer credit
```

7.2 Supplier

```
Purchase invoice: party_ledger +grand_total
Supplier payment: party_ledger -payment_amount
Purchase return/debit note: party_ledger -return_total
Supplier refund handling: post signed ledger entry based on settlement mode
```

```
supplier_balance = sum(supplier_ledger_signed_amount)
> 0 = payable, 0 = clear, < 0 = advance/debit credit
```

7.3 Invoice Payment Allocation

```
remaining_payment = payment_amount
for invoice in oldest_open_invoices:
    allocation = min(invoice.remaining_due, remaining_payment)
    create_payment_allocations_row
    update_invoice_remaining_due
    remaining_payment -= allocation
if remaining_payment > 0: keep as party credit/advance
```

- History should show invoice_status and party balance separately. Example: old invoice CLEARED, current party balance DUE because of a newer invoice.

8. Sales Return Flow

Step	UI / calculation action
Open	From Sales History select bill -> Return. Show original items, sold qty, already returned qty, returnable qty.
Validate	return_qty <= sold_qty - already_returned_qty and batch exists.
Calculate	Use original sale item snapshots. Do not use current GST/MRP/cost.
Stock	Post stock_ledger qty_in to original medicine_batch_id or return location.
GST	Reverse output GST using original item taxable/GST ratio.
Account	Post customer ledger -return_total. If cash refunded, post payment/refund ledger to consume credit.
Status	Mark sale PARTIAL_RETURN or RETURNED. Update sales_items.returned_qty.
Exports	Credit note, sales return register, GST return summary, profit reversal.

```
ratio = return_qty / original_qty
return_taxable = original_taxable_value * ratio
return_gst = original_gst_amount * ratio
return_line_total = original_line_total * ratio
profit_reversal = return_taxable - (return_qty * original_cost_rate_ex_gst_snapshot)
```

9. Purchase Return Flow

Step	UI / calculation action
Open	From Purchase History select purchase -> Return. Show purchased qty, free qty, sold/available qty, already returned qty.
Validate	Do not allow stock below zero unless policy allows negative stock with approval.
Calculate	Use original purchase item taxable/GST snapshots.
Stock	Post stock_ledger qty_out from medicine_batch_id.
GST	Reverse input GST for returned goods.
Account	Post supplier ledger -return_total because payable reduces. Settlement can be debit note, supplier credit, or refund.
Status	Mark purchase PARTIAL_RETURN or RETURNED. Update purchase_items.returned_qty.
Exports	Debit note, purchase return register, input GST reversal report.

```
ratio = return_qty / original_purchased_qty
return_taxable = original_taxable_value * ratio
return_input_gst = original_input_gst * ratio
return_line_total = original_line_total * ratio
```

10. UI Transformation

10.1 Navigation

Add ERP-style sidebar/top navigation grouped by module instead of only flat page buttons.

Group	Pages
Dashboard	Home, Today Summary, Alerts
Sales	New Sale, Sales Return, Sales History, Customer Ledger
Purchase	New Purchase, Purchase Return, Purchase History, Supplier Ledger
Inventory	Stock, Batches, Stock Movements, Expiry, Reorder, Locations
GST	Sales GST, Purchase GST, Return GST, Net GST, HSN Summary
Reports	Profit, Due, Payment, Stock Valuation, Audit, Export Center
Masters	Customers, Suppliers, Doctors, Medicines, Locations
Admin	Settings, Backup, Users, Numbering, Import

10.2 Sales UI

- Keep the first screen as actual billing, not a landing page.
- Item grid columns: Medicine, Batch, Expiry, Stock, Qty, Unit, MRP, Disc, Taxable, GST, Line Total, Cost/Profit warning optional.
- Summary panel: Subtotal inclusive, Item Discount, Bill Discount, Taxable Total, GST Total, Round Off, Grand Total, Credit Used, Cash, Online, Due, Customer Balance After.
- Actions: Save Bill, Print, Hold Bill, Clear, Use Credit, Add Return from History.

10.3 Purchase UI

- Header: supplier, GSTIN, bill no, supplier bill date, entry date, rate tax mode, payment terms.
- Item grid: Medicine, Batch, Expiry, Qty, Free Qty, Unit Factor, Rate, Discount, Taxable, GST, Line Total, MRP, Location.
- Summary: Taxable Total, Input GST, Discount, Round Off, Grand Total, Supplier Credit Used, Paid, Payable, Supplier Balance After.
- Actions: Save Purchase, Import Web JSON, Print Purchase, Create Purchase Return.

10.4 History UI

- Sales History and Purchase History should show status chips: OPEN, PARTIAL, CLEARED, PARTIAL_RETURN, RETURNED, VOID.
- Add side panel or detail drawer for item tax lines, payment allocations, ledger entries, stock movements, and audit trail.
- Right-click actions: View, Print, Export, Return, Receive/Pay, Void, Correction Entry. Avoid destructive delete for posted documents.

10.5 Return UI

- Return screens should open from original invoice to prevent wrong GST/cost source.
- Show returnable qty, previous returns, selected qty, reason, refund/credit/debit-note mode, and live GST reversal.
- Require reason for every return and void.

11. Export and Report Center

Report/export	Columns / purpose
Sales Register	Bill, customer, taxable, GST split, discount, round off, grand total, paid, due, status.
Sales Return Register	Return no, original bill, item, qty, taxable reversal, GST reversal, refund/credit mode.
Purchase Register	Supplier bill, supplier GSTIN, taxable, input GST, discount, grand total, paid, payable, status.
Purchase Return Register	Debit note, original purchase, item, qty, taxable/GST reversal, settlement.
GST Sales Report	Date range, HSN, GST rate, taxable, output CGST/SGST/IGST, sales returns adjusted.
GST Purchase Report	Supplier GSTIN, HSN, GST rate, taxable, input CGST/SGST/IGST, purchase returns adjusted.
Net GST Summary	Output GST - input GST by CGST/SGST/IGST with return adjustments.
Profit Report	Sale taxable - cost snapshot, adjusted by sales returns.
Stock Valuation	Batch stock * cost_rate_ex_gst and MRP value.
Stock Movement	Opening, purchases, sales, returns, adjustments, closing by batch.
Party Ledger	Customer/supplier ledger entries, balance after, invoice allocations.
Due Aging	0-30, 31-60, 61-90, 90+ due by customer/supplier.
Audit Log	Who changed/voided/corrected what and when.

- Keep current CSV/Excel/HTML export engine, but add report definitions from report_service and use one Export Center page.
- For PDF exports, generate print-ready HTML/PDF with pharmacy profile, date range, totals, and signatures where needed.

12. Current-to-ERP Implementation Phases

Phase	Work
1. Foundation	Add money/tax/ledger/stock services and write unit tests for formulas. No UI changes first.
2. Schema	Add new ERP tables: batches, ledgers, payments, allocations, returns, item GST fields, audit, sequences.
3. Purchase	Refactor Purchase save to use purchase_service, create supplier-linked batches and stock ledger.
4. Sales	Refactor Sales save to use sales_service, freeze GST/cost snapshots and post customer ledger.
5. Histories	Rebuild histories to read invoice snapshots, ledger allocations, returns, and status.
6. Returns	Add Sales Return and Purchase Return pages and posting services.
7. Reports	Add GST, profit, return, stock valuation, ledger, and aging reports.
8. UI polish	ERP navigation, status chips, detail drawers, export center, role permissions.
9. Cutover	For new users, use ERP tables as source of truth. Keep old columns only for backward compatibility if needed.

13. Validation Test Matrix

Test	Expected result
Sale paid exact	Invoice CLEARED, customer balance unchanged after payment, stock out posted.
Sale partial paid	Invoice PARTIAL/OPEN, customer balance positive due.
Sale overpaid	Invoice CLEARED, customer credit negative balance. Next sale can use credit.
Sales return partial	Stock in, GST output reversal, customer balance reduced, profit reversed.
Purchase paid exact	Purchase CLEARED, supplier balance zero, stock in posted.
Purchase partial paid	Supplier payable positive, purchase PARTIAL/OPEN.
Supplier advance	Supplier balance negative advance, next purchase uses credit.
Purchase return partial	Stock out, input GST reversal, supplier payable reduced.
Mixed GST invoice	Rate-wise GST totals match line sums and report totals.
Bill discount before GST	Discount allocated line-wise before GST; HSN/rate reports correct.
Void bill	Reversal ledger and stock entries created; reports handle status correctly.
History clarity	Old cleared invoices stay cleared after new due invoice; party balance shows due separately.

14. Final Recommendation

To make the current version a complete ERP, do not only add Sales Return and Purchase Return screens on top of the current tables. First create the calculation, ledger, stock, and tax foundation. Then connect Sales and Purchase entry to that foundation, then rebuild both history pages and exports. Returns become clean only when original item GST, discount, batch, supplier, and cost snapshots are already stored. This approach gives accurate GST filing, accurate profit, clear account histories, and ERP-grade auditability.